

Text Processing & Feature Engineering Report

1. Objective

The objective of this analysis was to preprocess textual review data and transform it into numerical representations suitable for machine learning tasks such as sentiment analysis.

2. Dataset Overview

The dataset consists of 50 product reviews labeled as positive (1) or negative (0), covering product quality, delivery, and usability.

3. Data Preprocessing

Text was converted to lowercase and processed using SpaCy for tokenization, stopword removal, and punctuation cleaning.

Original Text	Clean Tokens
The product is okay, but shipping took too long.	product, okay, shipping, took, long

4. Feature Engineering Techniques

Bag of Words captures frequency, One-Hot Encoding captures presence, and TF-IDF assigns importance weights to words.

5. Matrix Characteristics

Representation	Shape	Description
BoW	(50, 118)	Frequency-based
One-Hot	(50, 118)	Binary presence
TF-IDF	(50, 118)	Weighted importance

6. Sparsity Analysis

All representations showed ~96.88% sparsity, meaning most values are zero.

7. Observations

- 1 Reviews focus on product quality, performance, and delivery.
- 2 Positive words include excellent, amazing, great.
- 3 Negative words include poor, terrible, defective.

8. Conclusions

TF-IDF provides better feature representation, but all methods suffer from sparsity and lack semantic understanding.

9. Recommendations

- 1 Use word embeddings like Word2Vec or GloVe.
- 2 Use transformer models like BERT.
- 3 Apply dimensionality reduction techniques.